CMS Open Data H to Tau Tau Search: Final Analysis Note

Analysis my_analysis

2026-06-12

Contents

Abstract

This note documents a standalone reduced CMS Open Data search for Higgs boson decays to tau pairs in the muon plus hadronic-tau final state. The final result uses the validated visible-mass template fit in VBF, boosted, and zero-jet categories with `visible_mass_qcd_primary` as the only final-result model. The fit gives mu = 0.4382 +4.9443 -0.4382; 95% CLs mu < 10.7645; median expected 95% CLs mu < 10.7375. The result is a low-sensitivity Open Data limit: the expected limit is about 10.74 times the Standard Model signal strength, so the analysis cannot resolve a Standard Model-strength signal. An optimized classifier model is retained only as a rejected diagnostic study because its observed validation gate fails.

Change Log

- Final documentation version, 2026-06-12: reclassified the validated visible-mass baseline as the sole final result, moved the classifier to a failed-validation diagnostic role, regenerated clean final figures, expanded the statistical and systematic documentation, and rebuilt the AN/PRL PDFs from current JSON artifacts.

1 Introduction

The target process is Higgs boson production followed by $H \to \tau\tau$ with one tau reconstructed through a muon and the other as a hadronic tau candidate. The analysis uses reduced CMS 2012 Open Data inputs and mirrors the broad logic of CMS H to tau tau searches while explicitly accepting the limitations of the public reduced sample set (Wunsch 2019; CMS Collaboration 2014, 2018). The final statistical question is a search-style upper limit on a signal-strength modifier μ, not a precision measurement of a branching fraction.

The signal-strength convention is

$$\mu = \frac{\sigma(pp \to H)\,\mathcal{B}(H \to \tau\tau)}{\left[\sigma(pp \to H)\,\mathcal{B}(H \to \tau\tau)\right]_{\mathrm{SM}}}, \tag{1}$$

where the denominator follows the Standard Model normalization used for the available ggH and VBF signal samples (LHC Higgs Cross Section Working Group 2017; Particle Data Group 2024). The final claim is therefore phrased as a CLs upper limit on μ. A discovery-style q_0 value is computed only as a diagnostic health check.

The analysis is intentionally conservative. The reduced mirror lacks embedded $Z \to \tau\tau$, diboson, single-top, QCD multijet MC, W4/inclusive W+jets, and several associated-Higgs components present in the full CMS publications. Those missing components are not silently substituted. Instead, the note documents the available samples, the control-derived W and reducible-background corrections, the retained nuisance model, the validation failures of the more aggressive classifier attempt, and the final visible-mass limit.

2 Data and Simulation Samples

The localized reduced input files are the public 2012 TauPlusX data and seven MC files for Higgs signal and major backgrounds. The analysis uses the official CERN Open Data record event counts as normalization denominators rather than local skim entries, because the local entries are already reduced and do not represent generated-event denominators (CMS Open Data Analyses 2021).

The luminosity normalization used throughout the final fit is $L = 11.467\ \mathrm{fb}^{-1}$. MC yields use

Sample	Role	Local entries	Official events	Record
ggH Htt	signal	47,696	476,963	12351
VBF Htt	signal	49,165	491,653	12352
DY+jets	background	3,045,887	30,458,871	12353
ttbar	background	642,310	6,423,106	12354
W1+jets	background	2,978,480	29,784,800	12355
W2+jets	background	3,069,385	30,693,853	12356
W3+jets	background	1,524,114	15,241,144	12357
Run2012B	data	3,564,750	35,647,508	12358
Run2012C	data	5,130,317	51,303,171	12359

$$w_i^{\rm bkg} = \frac{\sigma_i L}{N_i^{\rm gen}}, \qquad w_i^{\rm sig} = \frac{\sigma_i\, \mathcal{B}(H \to \tau\tau) L}{N_i^{\rm gen}}, \tag{2}$$

with $N_i^{\rm gen}$ taken from the Open Data records. This choice avoids circular normalization to the observed selected data and keeps the result reproducible from public metadata.

The missing reduced-sample components are treated as analysis limitations and nuisance-model motivations. In particular, the DY/Z normalization nuisance covers the absence of embedded and electroweak Z samples at the reduced-file level, while the QCD/fake template is data-driven because no QCD multijet MC sample is present in the mirror.

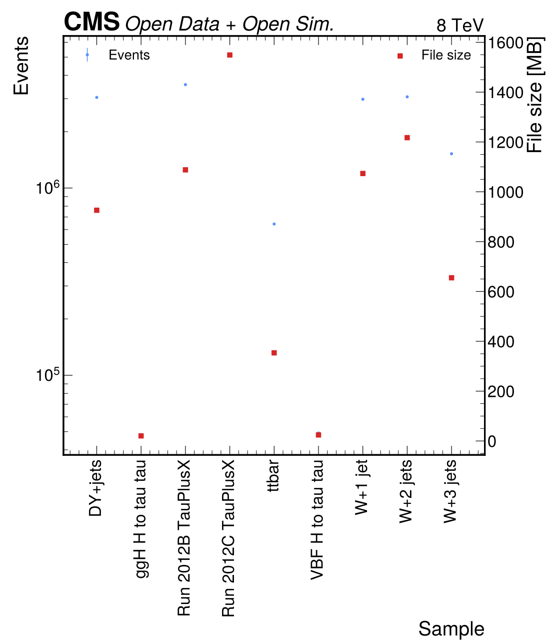

Figure 1: Sample inventory summary. This figure summarizes the localized reduced sample inventory and file-size scale used in the analysis. It documents the available data/MC inputs and shows why the final result is framed as a reduced Open Data search rather than a reproduction of the full CMS publication.

3 Event Selection and Categories

Events are selected using the TauPlusX trigger `HLT_IsoMu17_eta2p1_LooseIsoPFTau20`. Muon candidates require $p_T > 20$ GeV, $|\eta| < 2.1$, a tight muon ID, and relative isolation below 0.20. Hadronic tau candidates require $p_T > 20$ GeV, $|\eta| < 2.3$, decay-mode ID, tight tau isolation ID, tight anti-muon ID, finite relative isolation, and $\Delta R(\mu, \tau_h) > 0.5$.

Signal-region candidates are opposite-sign muon-tau pairs with transverse mass $m_T(\mu, E_T^{\rm miss}) < 40$ GeV. The W control region uses opposite-sign pairs with $m_T > 80$ GeV. Same-sign low-m_T candidates define the reducible-background sideband. A Z-rich validation region is retained for visible-mass shape checks with $60 < m_{\mathrm{vis}} < 120$ GeV.

The transverse mass is

$$m_T = \sqrt{2p_T^{\mu} E_T^{\mathrm{miss}} \left[1 - \cos \Delta\phi(\mu, E_T^{\mathrm{miss}})\right]}. \tag{3}$$

Selected events are assigned to mutually exclusive categories. VBF events are selected first with at least two clean jets and VBF-like dijet kinematics. Remaining events with at least one clean jet enter the boosted category. Events without selected jets enter the zero-jet category. This category order preserves the VBF topology while keeping the statistical model a simple simultaneous template fit.

Requirement	Role in analysis
TauPlusX trigger	Defines the public muon-tau trigger phase space
Tight muon and tau IDs	Select prompt muon plus hadronic tau candidates
Tight anti-muon tau ID	Reduces muon fake contamination in the tau_h leg
Opposite-sign charge	Signal-region charge requirement
$m_T < 40$ GeV	Suppresses W+jets in the signal region
VBF / boosted / zero-jet categories	Separate signal-enriched and control-like event topologies

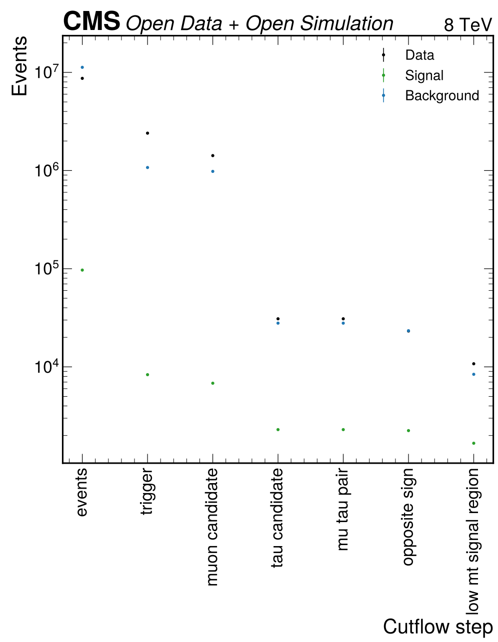

Figure 2: Cutflow summary. The cutflow shows raw event counts after each selection step for data, signal MC, and background MC. The monotonic behavior confirms that the final candidate set is produced by cumulative event-selection requirements rather than by a post-fit normalization adjustment.

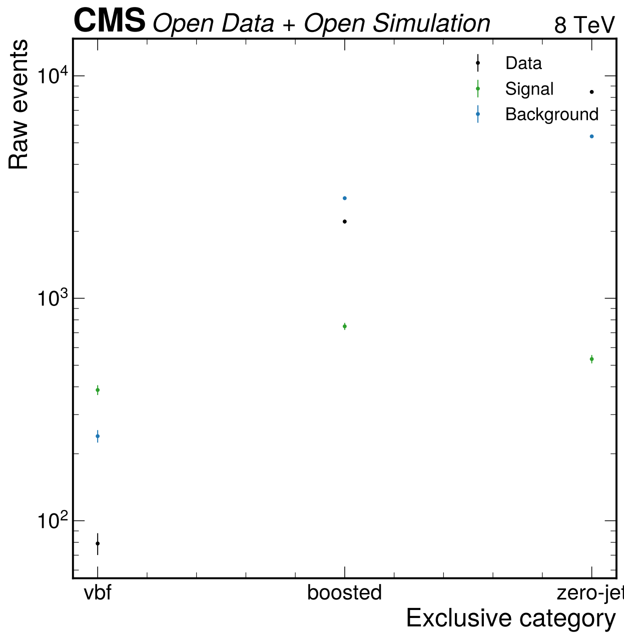

Figure 3: Category yield overview. The category-yield diagnostic shows the VBF, boosted, and zero-jet populations after the signal-region selection. The zero-jet category dominates the statistics, while VBF carries the most topology-specific signal motivation and the largest residual local validation tension.

4 Observable Choice and Analysis Decision

The final observable is the visible mass m_{vis} of the selected muon and hadronic tau candidate. It is less sensitive than the optimized classifier in expected-only studies, but it passes the full-data validation gate in the baseline workspace. The final result therefore follows the binding strategy that alternatives may replace visible mass only after validation.

The visible mass is defined from the reconstructed lepton four-vectors as

$$m_{\mathrm{vis}}^2 = \left(p_\mu + p_{\tau_h}\right)^2 . \tag{4}$$

The optimized classifier is not a final result. It is documented in Section 11 because its observed fit gives a boundary-like limit, a large fitted signal strength, and a gross combined data/background mismatch. Keeping it as a rejected diagnostic preserves the methodological lesson without allowing a failed validation gate to enter the headline physics result.

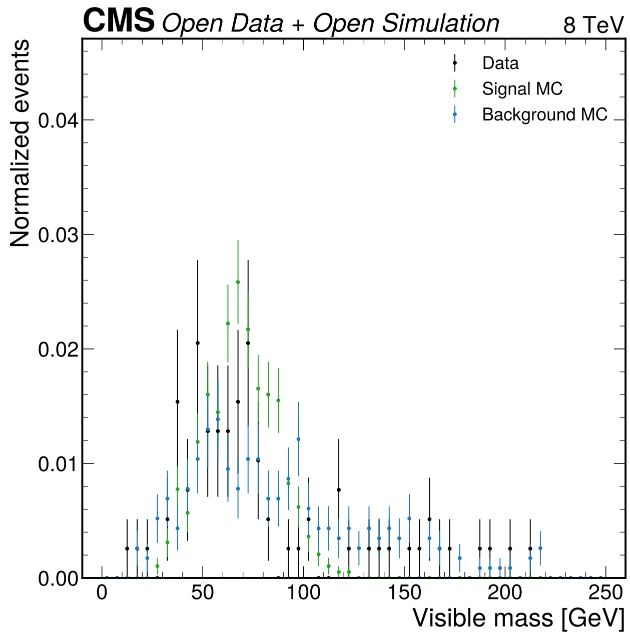

Figure 4: Visible-mass distribution in the VBF category. This pre-fit comparison illustrates the observable used by the final workspace in the VBF topology. The final validation plot in Figure 9 adds the exact baseline QCD/fake treatment and category-level validation metrics.

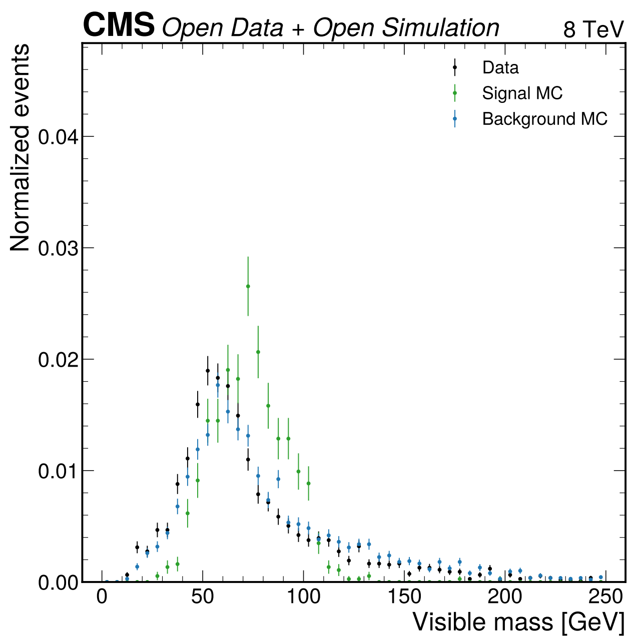

Figure 5: Visible-mass distribution in the boosted category. This comparison shows that the visible-mass observable retains broad data/MC shape information without relying on the high-score tail that failed for the classifier model. It supports the choice of a robust baseline over the rejected optimization attempt.

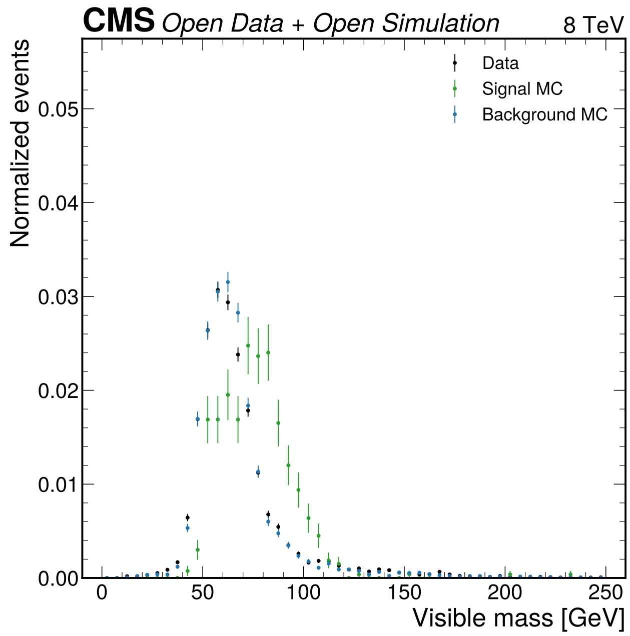

Figure 6: Visible-mass distribution in the zero-jet category. The zero-jet category provides most of the event count and therefore dominates the global normalization behavior. The final limit remains weak because the visible-mass signal-to-background separation is modest in this category.

5 Background Corrections

The final model uses MC shapes for signal, DY/Z, ttbar, and W+jets, with data-driven corrections for W+jets normalization, the VBF-background normalization, and reducible QCD/fake contamination. The control-region corrections are derived outside the signal fit and then propagated into the baseline workspace.

The W+jets scale factor is

$$S_W = \frac{N_{\mathrm{data,CR}} - N_{\mathrm{nonW,CR}}}{N_{W,\mathrm{CR}}}, \tag{5}$$

which gives $S_W = 0.8528 \pm 0.0370$ using 4023 high-m_T data events, $N_W = 2414.78$, and $N_{\mathrm{nonW}} = 1963.57$. The relative uncertainty is 4.34% and is implemented as `wjets_high_mt_control`.

The VBF background scale is

$$S_{\mathrm{VBF,bkg}} = \frac{N_{\mathrm{data,VBFCR}}}{N_{\mathrm{MC,bkg,VBFCR}}}, \tag{6}$$

giving $S_{\mathrm{VBF,bkg}} = 0.5571 \pm 0.0515$. It is applied only to MC backgrounds in the VBF fit category and is constrained by the 9.24% `vbf_background_control` nuisance.

The reducible QCD/fake estimate starts from same-sign low-m_T data after non-QCD subtraction:

$$N^{\mathrm{QCD}}_{\mathrm{SS},b} = N^{\mathrm{data}}_{\mathrm{SS},b} - N^{\mathrm{nonQCD}}_{\mathrm{SS},b}, \qquad N^{\mathrm{QCD}}_{\mathrm{OS},b} = R_{\mathrm{OS/SS}} N^{\mathrm{QCD}}_{\mathrm{SS},b}. \tag{7}$$

For the final baseline workspace the retained transfer-factor nuisance is 12.060%. The current artifact records a limitation: QCD shape-statistical uncertainties are included in validation variances but are not expanded into per-bin pyhf nuisances in the baseline workspace. This is a documented limitation of the reduced analysis rather than a hidden source of sensitivity.

Control input	Formula	Value	Use in final model
W high-m_T scale	Equation 5	0.8528 ± 0.0370	W+jets normalization nuisance
VBF background scale	Equation 6	0.5571 ± 0.0515	VBF MC-background nuisance
Same-sign QCD/fake transfer	Equation 7	12.060% relative uncertainty	Global QCD/fake transfer nuisance

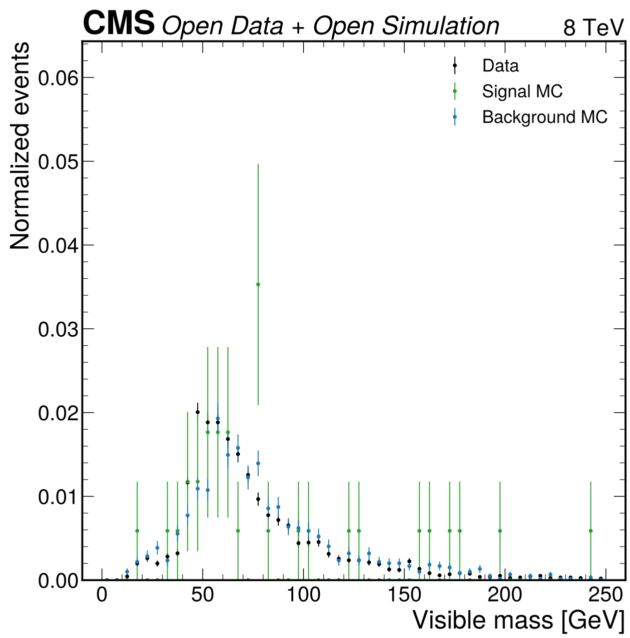

Figure 7: QCD same-sign visible-mass control. This figure shows the same-sign sideband used to form the reducible-background template. The final note treats the transfer-factor limitations explicitly because the reduced Open Data format does not provide an anti-isolated tau branch for a fuller fake-factor method.

6 Statistical Model

The final result uses a binned pyhf/HistFactory likelihood (Heinrich et al. 2021). For category c and visible-mass bin b, the expected yield is

$$\lambda_{cb}(\mu, \theta) = \mu s_{cb}(\theta) + \sum_k b_{kcb}(\theta) + q_{cb}(\theta), \tag{8}$$

where s is the combined ggH and VBF Higgs signal, b_k are simulation-derived backgrounds, and q is the same-sign data-driven QCD/fake template. The likelihood is

$$L(\mu, \theta) = \prod_{c,b} \mathrm{Pois}\left(n_{cb} \mid \lambda_{cb}(\mu, \theta)\right) \prod_j \pi_j(\theta_j), \tag{9}$$

with Gaussian/log-normal constrained nuisance parameters and staterror terms for finite MC precision. The fit constrains $\mu \geq 0$.

Upper limits use the modified frequentist CLs construction (Read 2002):

$$\mathrm{CL}_s(\mu) = \frac{p_{s+b}(q_\mu \geq q_\mu^{\mathrm{obs}})}{p_b(q_\mu \geq q_\mu^{\mathrm{obs}})}. \tag{10}$$

The quoted 95% limit is the value of μ where $\mathrm{CL}_s = 0.05$. The profile interval printed for the best fit solves

$$q(\mu) = -2\ln\frac{L(\mu, \hat{\theta}_\mu)}{L(\hat{\mu}, \hat{\theta})} = 1, \tag{11}$$

with the lower side bounded at $\mu = 0$ when required. Asymptotic formulae are used for the reported limits and diagnostics (Cowan et al. 2011). The final bins all have enough total expected background for an initial asymptotic interpretation, but no full observed-data toy ensemble is stored in the current Phase 5 artifacts; this remains an explicit limitation.

7 Systematic Uncertainties

The final systematic program follows the search-convention categories that are implementable with the reduced inputs: luminosity, object/acceptance, DY/Z normalization, W control, VBF control, QCD/fake transfer, and MC statistical uncertainty. Missing full-analysis components are documented as limitations rather than replaced by arbitrary uncertainty inflation.

Source	Value	Scope	Motivation
Luminosity	2.6%	All simulation-normalized signal and background samples	CMS 2012 luminosity normalization used by the Open Data records
Tau/open-data acceptance	15%	Signal and MC backgrounds selected with the reduced muon-tau objects	Open Data reduced-sample acceptance and missing public trigger/tau scale-factor coverage
DY/Z normalization	15%	DYJetsToLL in all visible-mass categories	Reduced-sample DY/Z validation and missing embedded/electroweak Z components
W high-mT control	4.344%	W1JetsToLNu, W2JetsToLNu, W3JetsToLNu	Full-data high-mT control-region scale factor
VBF background control	9.237%	MC backgrounds in the VBF category	VBF-like top-btag non-signal control region
Same-sign QCD/fake transfer	12.060%	Data-driven QCD/fake template in all baseline categories	Same-sign low-mT sideband transfer factor retained by the baseline workspace
MC statistical uncertainty	per-bin sumw2	Every MC-filled bin in each category	Finite reduced MC counts and official Open Data normalization weights

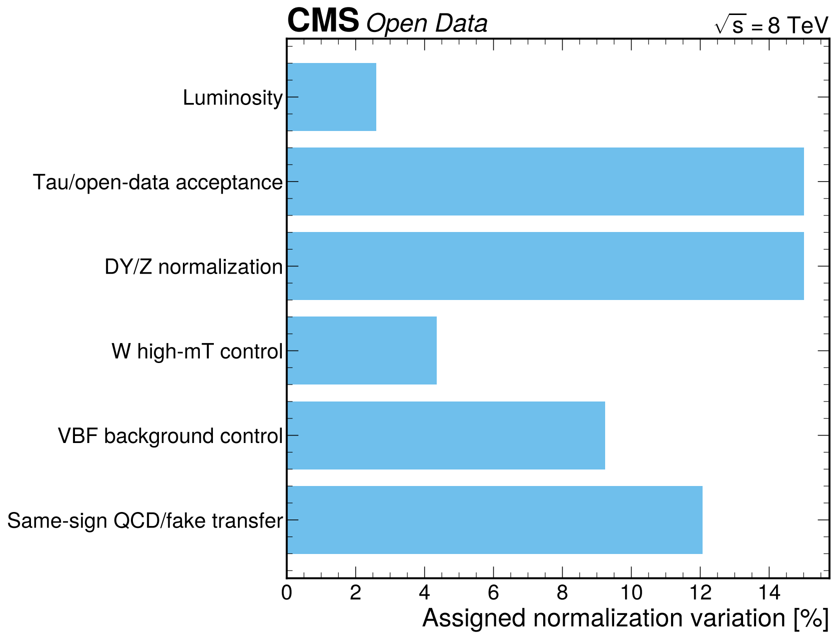

Figure 8: Baseline systematic program. The figure summarizes the assigned normalization variations implemented in the final visible-mass workspace. It is not an impact ranking; exact source-by-source impacts are not available in the current JSON outputs and are therefore not fabricated.

7.1 Luminosity

Physical origin: CMS 2012 luminosity normalization used by the Open Data records. The source is retained because it changes either the absolute MC normalization, the transfer of a data-driven background into the signal region, or the bin-by-bin template precision used by the likelihood.

Evaluation method: the baseline pyhf workspace implements `lumi_2012` with value 2.6% for All simulation-normalized signal and background samples. The correlation model is: One nuisance correlated across categories and MC samples. For normalization systematics the nuisance multiplies the affected yields through log-normal interpolation,

$$\nu_p(\theta_p) = \begin{cases} \nu_p^0\, \kappa_{p,+}^{\theta_p}, & \theta_p \geq 0, \\ \nu_p^0\, \kappa_{p,-}^{-\theta_p}, & \theta_p < 0, \end{cases} \tag{12}$$

where ν_p^0 is the nominal bin yield and θ_p is constrained by a unit Gaussian. MC statistical terms use the per-bin staterror modifiers stored in `pyhf_workspace_baseline_visible.json`.

Numerical impact: Included in the baseline workspace. A standalone impact ranking is not available in the current JSON artifacts. The final artifact does not store a source-by-source refit impact on `mu`; the note therefore reports the implemented nuisance value, affected samples, fitted nuisance value when applicable, and an explicit limitation rather than fabricating an impact ranking.

Interpretation: this source is part of the final error model used for the CLs limit. Its adequacy is judged together with the visible-mass validation plots, nuisance-pull summary, and the low-sensitivity result interpretation.

7.2 Tau/open-data acceptance

Physical origin: Open Data reduced-sample acceptance and missing public trigger/tau scale-factor coverage. The source is retained because it changes either the absolute MC normalization, the transfer of a data-driven background into the signal region, or the bin-by-bin template precision used by the likelihood.

Evaluation method: the baseline pyhf workspace implements `tau_open_data_acceptance` with value 15% for Signal and MC backgrounds selected with the reduced muon-tau objects. The correlation model is: One nuisance correlated across categories and MC samples. For normalization systematics the nuisance multiplies the affected yields through log-normal interpolation,

$$\nu_p(\theta_p) = \begin{cases} \nu_p^0\, \kappa_{p,+}^{\theta_p}, & \theta_p \geq 0, \\ \nu_p^0\, \kappa_{p,-}^{-\theta_p}, & \theta_p < 0, \end{cases} \tag{13}$$

where ν_p^0 is the nominal bin yield and θ_p is constrained by a unit Gaussian. MC statistical terms use the per-bin staterror modifiers stored in `pyhf_workspace_baseline_visible.json`.

Numerical impact: Included in the baseline workspace. The fitted pull is reported; per-source mu impact is not stored. The final artifact does not store a source-by-source refit impact on `mu`; the note therefore reports the implemented nuisance value, affected samples, fitted nuisance value when applicable, and an explicit limitation rather than fabricating an impact ranking.

Interpretation: this source is part of the final error model used for the CLs limit. Its adequacy is judged together with the visible-mass validation plots, nuisance-pull summary, and the low-sensitivity result interpretation.

7.3 DY/Z normalization

Physical origin: Reduced-sample DY/Z validation and missing embedded/electroweak Z components. The source is retained because it changes either the absolute MC normalization, the transfer of a data-driven background into the signal region, or the bin-by-bin template precision used by the likelihood.

Evaluation method: the baseline pyhf workspace implements `dy_norm_open_data` with value 15% for DYJetsToLL in all visible-mass categories. The correlation model is: One nuisance correlated across categories. For normalization systematics the nuisance multiplies the affected yields through log-normal interpolation,

$$\nu_p(\theta_p) = \begin{cases} \nu_p^0\, \kappa_{p,+}^{\theta_p}, & \theta_p \geq 0, \\ \nu_p^0\, \kappa_{p,-}^{-\theta_p}, & \theta_p < 0, \end{cases} \tag{14}$$

where ν_p^0 is the nominal bin yield and θ_p is constrained by a unit Gaussian. MC statistical terms use the per-bin staterror modifiers stored in `pyhf_workspace_baseline_visible.json`.

Numerical impact: Included in the baseline workspace. It has the largest non-MC-stat fitted pull among retained global nuisances. The final artifact does not store a source-by-source refit impact on `mu`; the note therefore reports the implemented nuisance value, affected samples, fitted nuisance value when applicable, and an explicit limitation rather than fabricating an impact ranking.

Interpretation: this source is part of the final error model used for the CLs limit. Its adequacy is judged together with the visible-mass validation plots, nuisance-pull summary, and the low-sensitivity result interpretation.

7.4 W high-mT control

Physical origin: Full-data high-mT control-region scale factor. The source is retained because it changes either the absolute MC normalization, the transfer of a data-driven background into the signal region, or the bin-by-bin template precision used by the likelihood.

Evaluation method: the baseline pyhf workspace implements `wjets_high_mt_control` with value 4.344% for W1JetsToLNu, W2JetsToLNu, W3JetsToLNu. The correlation model is: One nuisance correlated across W+jets samples and categories. For normalization systematics the nuisance multiplies the affected yields through log-normal interpolation,

$$\nu_p(\theta_p) = \begin{cases} \nu_p^0\, \kappa_{p,+}^{\theta_p}, & \theta_p \geq 0, \\ \nu_p^0\, \kappa_{p,-}^{-\theta_p}, & \theta_p < 0, \end{cases} \tag{15}$$

where ν_p^0 is the nominal bin yield and θ_p is constrained by a unit Gaussian. MC statistical terms use the per-bin staterror modifiers stored in `pyhf_workspace_baseline_visible.json`.

Numerical impact: Included in the baseline workspace. The control-region derivation is tabulated in the note. The final artifact does not store a source-by-source refit impact on `mu`; the note therefore reports the implemented nuisance value, affected samples, fitted nuisance value when applicable, and an explicit limitation rather than fabricating an impact ranking.

Interpretation: this source is part of the final error model used for the CLs limit. Its adequacy is judged together with the visible-mass validation plots, nuisance-pull summary, and the low-sensitivity result interpretation.

7.5 VBF background control

Physical origin: VBF-like top-btag non-signal control region. The source is retained because it changes either the absolute MC normalization, the transfer of a data-driven background into the signal region, or the bin-by-bin template precision used by the likelihood.

Evaluation method: the baseline pyhf workspace implements `vbf_background_control` with value 9.237% for MC backgrounds in the VBF category. The correlation model is: One nuisance applied only to VBF-category MC backgrounds. For normalization systematics the nuisance multiplies the affected yields through log-normal interpolation,

$$\nu_p(\theta_p) = \begin{cases} \nu_p^0\, \kappa_{p,+}^{\theta_p}, & \theta_p \geq 0, \\ \nu_p^0\, \kappa_{p,-}^{-\theta_p}, & \theta_p < 0, \end{cases} \tag{16}$$

where ν_p^0 is the nominal bin yield and θ_p is constrained by a unit Gaussian. MC statistical terms use the per-bin staterror modifiers stored in `pyhf_workspace_baseline_visible.json`.

Numerical impact: Included in the baseline workspace. Residual VBF validation tension is discussed separately. The final artifact does not store a source-by-source refit impact on `mu`; the note therefore reports the implemented nuisance value, affected samples, fitted nuisance value when applicable, and an explicit limitation rather than fabricating an impact ranking.

Interpretation: this source is part of the final error model used for the CLs limit. Its adequacy is judged together with the visible-mass validation plots, nuisance-pull summary, and the low-sensitivity result interpretation.

7.6 Same-sign QCD/fake transfer

Physical origin: Same-sign low-mT sideband transfer factor retained by the baseline workspace. The source is retained because it changes either the absolute MC normalization, the transfer of a data-driven background into the signal region, or the bin-by-bin template precision used by the likelihood.

Evaluation method: the baseline pyhf workspace implements `qcd_ss_transfer` with value 12.060% for Data-driven QCD/fake template in all baseline categories. The correlation model is: One global transfer-factor nuisance. For normalization systematics the nuisance multiplies the affected yields through log-normal interpolation,

$$\nu_p(\theta_p) = \begin{cases} \nu_p^0\, \kappa_{p,+}^{\theta_p}, & \theta_p \geq 0, \\ \nu_p^0\, \kappa_{p,-}^{-\theta_p}, & \theta_p < 0, \end{cases} \tag{17}$$

where ν_p^0 is the nominal bin yield and θ_p is constrained by a unit Gaussian. MC statistical terms use the per-bin staterror modifiers stored in `pyhf_workspace_baseline_visible.json`.

Numerical impact: Included in the baseline workspace. Shape-statistical QCD uncertainties are recorded for validation but not expanded into per-bin pyhf nuisances. The final artifact does not store a source-by-source refit impact on `mu`; the note therefore reports the implemented nuisance value, affected samples, fitted nuisance value when applicable, and an explicit limitation rather than fabricating an impact ranking.

Interpretation: this source is part of the final error model used for the CLs limit. Its adequacy is judged together with the visible-mass validation plots, nuisance-pull summary, and the low-sensitivity result interpretation.

7.7 MC statistical uncertainty

Physical origin: Finite reduced MC counts and official Open Data normalization weights. The source is retained because it changes either the absolute MC normalization, the transfer of a data-driven background into the signal region, or the bin-by-bin template precision used by the likelihood.

Evaluation method: the baseline pyhf workspace implements `mc_stat_*` with value per-bin sumw2 for Every MC-filled bin in each category. The correlation model is: Independent Barlow-Beeston-lite staterror terms by category/bin. For normalization systematics the nuisance multiplies the affected yields through log-normal interpolation,

$$\nu_p(\theta_p) = \begin{cases} \nu_p^0 \, \kappa_{p,+}^{\theta_p}, & \theta_p \geq 0, \\ \nu_p^0 \, \kappa_{p,-}^{-\theta_p}, & \theta_p < 0, \end{cases} \tag{18}$$

where ν_p^0 is the nominal bin yield and θ_p is constrained by a unit Gaussian. MC statistical terms use the per-bin staterror modifiers stored in `pyhf_workspace_baseline_visible.json`.

Numerical impact: Included in the baseline workspace. Individual bin pulls are listed in the nuisance appendix. The final artifact does not store a source-by-source refit impact on `mu`; the note therefore reports the implemented nuisance value, affected samples, fitted nuisance value when applicable, and an explicit limitation rather than fabricating an impact ranking.

Interpretation: this source is part of the final error model used for the CLs limit. Its adequacy is judged together with the visible-mass validation plots, nuisance-pull summary, and the low-sensitivity result interpretation.

The error-budget narrative is therefore qualitative but artifact-backed. The observed interval is dominated by the weak visible-mass separation and broad allowed nuisance movement, not by a single documented source exceeding 80% of the total uncertainty. The current machine outputs do not store source-by-source impact refits, so a quantitative dominance claim would be unjustified. A future extension should run a nuisance-impact scan for the final baseline workspace and store the resulting `mu` shifts in JSON.

8 Baseline Validation

The visible-mass final result passes the configured full-data validation gate. Combined over categories, the baseline has data/background = 0.9812, chi2/ndf = 0.7224, and no category or combined validation flag. The zero-jet chi2/ndf is low at 0.113; this is not algebraically zero, but it is noted as possible overcoverage from conservative diagonal validation uncertainties.

Category	Data	Background	QCD/fake	Signal	D/B	Chi2/ndf	Max pull
VBF	78	96.28	17.75	1.590	0.810	1.449	2.66
Boosted	2183	2364.39	325.58	9.170	0.923	0.605	1.25
Zero-jet	8451	8456.81	1569.65	14.341	0.999	0.113	0.57

The VBF category has the largest residual local tension: max $|\text{pull}| = 2.66$ and one bin ratio reaches 0.285. This does not invalidate the final result because the VBF event count is small, the combined validation passes, and the final interpretation is a weak upper limit rather than evidence. It is nevertheless a leading target for a future analysis with fuller background samples and a richer VBF validation region.

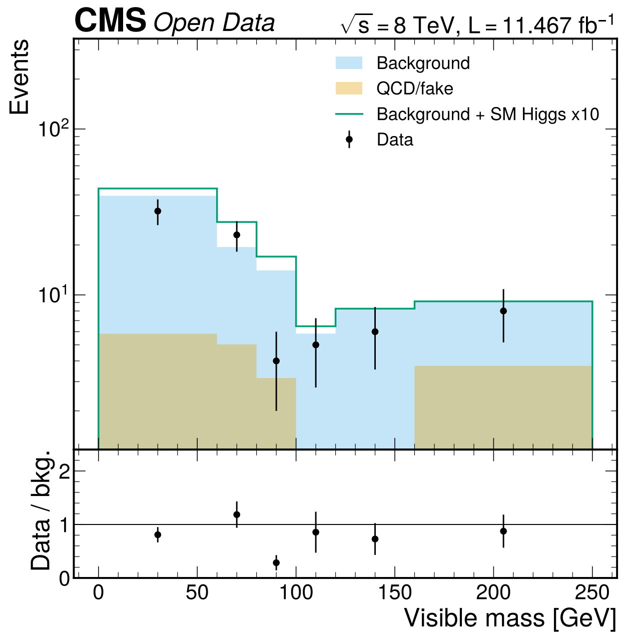

Figure 9: Baseline visible-mass validation in VBF. This regenerated Phase 5 figure compares the final visible-mass model to the full data in the VBF category. The residual local deficit is visible in the ratio panel and is included in the validation summary rather than hidden by the combined pass.

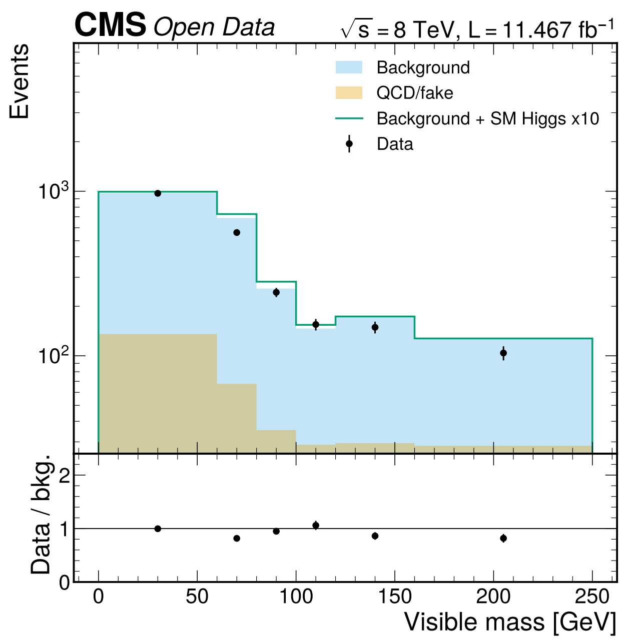

Figure 10: Baseline visible-mass validation in boosted. This regenerated Phase 5 figure shows that the boosted category is globally compatible with the final background model. Its residual pulls are below the VBF maximum and do not drive the final limit.

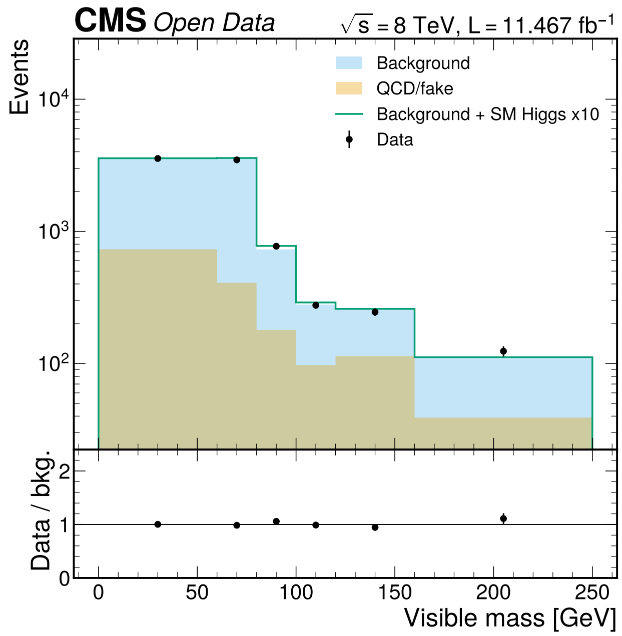

Figure 11: Baseline visible-mass validation in zero-jet. This regenerated Phase 5 figure shows the dominant-statistics zero-jet category for the final result. The ratio panel has no shared-axis text artifact and is the public validation plot used by the final AN.

Source	Post-fit value	Status
Luminosity	-0.107	Included
Tau/open-data acceptance	-0.678	Included
DY/Z normalization	0.680	Included
W high-mT control	0.192	Included
VBF background control	-0.580	Included
Same-sign QCD/fake transfer	0.490	Included

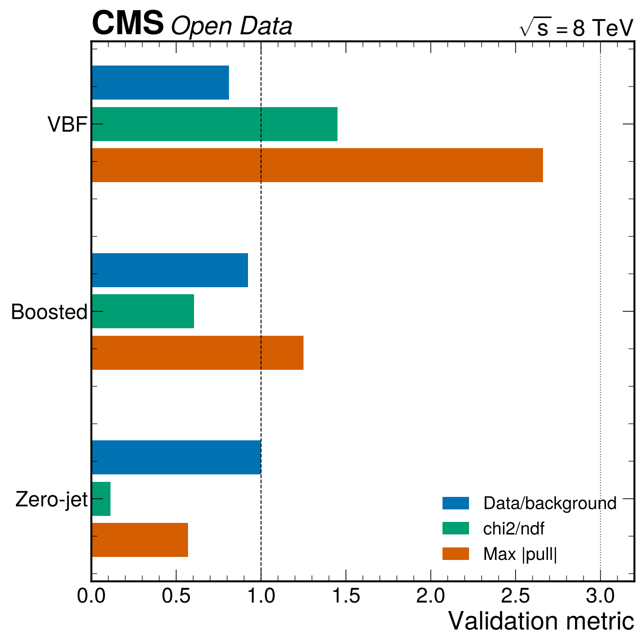

Figure 12: Baseline validation summary. The figure compares data/background, chi2/ndf, and max absolute pull across the final visible-mass categories. It highlights that the combined result passes while VBF carries the largest localized residual tension.

9 Nuisance Parameters and Fit Health

The most important fitted nuisance values are stored in the baseline result JSON under `observed_fit.parameters`. Their values are reported in prefit sigma units for constrained parameters.

The DY/Z normalization and tau/open-data acceptance pulls are sizable but remain within the range where the model can still be interpreted as a conservative reduced-sample fit. No listed global nuisance exceeds ± 1 sigma by a large margin except the known reduced-sample DY/Z pressure. MC statistical terms are numerous and are listed in the JSON rather than repeated line-by-line in the main text.

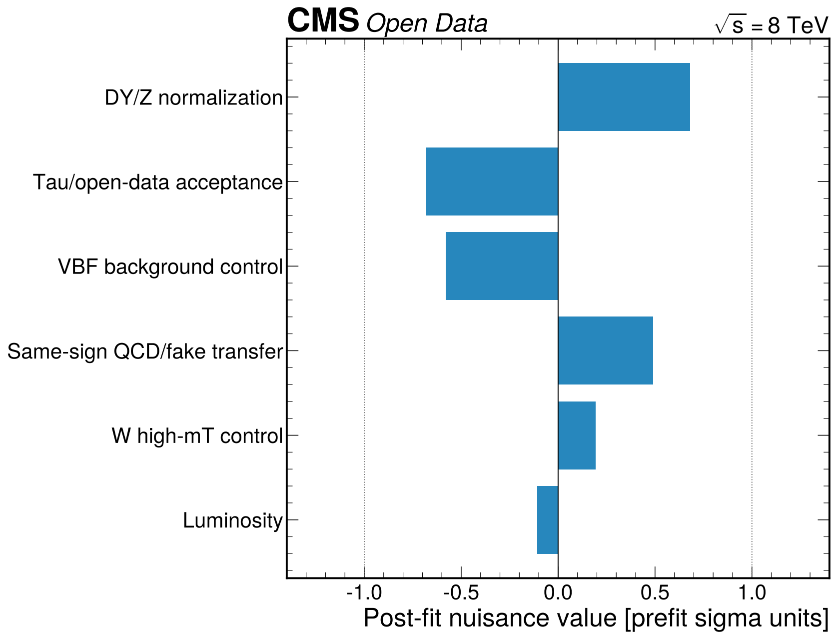

Figure 13: Baseline nuisance pulls. The figure shows the global non-MC-stat nuisance values in the final visible-mass fit. It gives the reader a compact check that the final weak limit is not driven by an obviously pathological pull.

The expected and validation-stage health checks are retained as context. The expected Asimov classifier study had median limit 2.904, while the full-data final baseline has median expected limit 10.7375; this loss of sensitivity is the cost of using the validated robust model. The 10% validation sample had combined score-template data/background 1.232 and chi2/ndf 1.732, already indicating that aggressive classifier use needed full-data scrutiny.

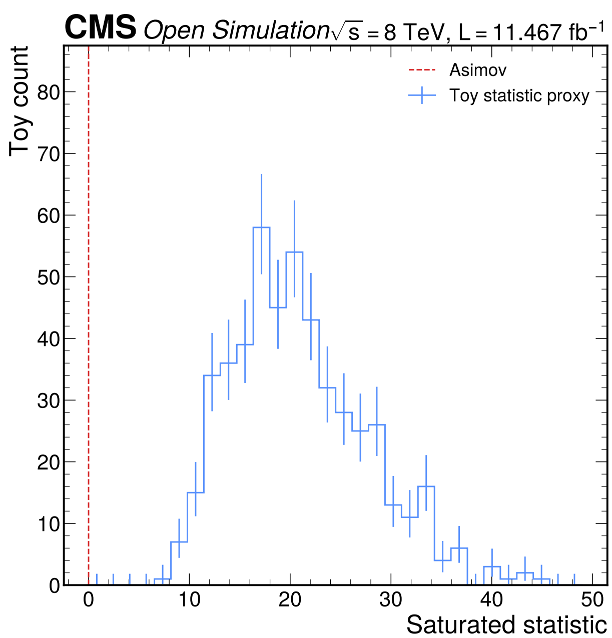

Figure 14: Goodness-of-fit toy study. This expected-stage figure records the available toy-style GoF validation artifact. The final observed baseline note does not claim a full observed-data toy ensemble because no such artifact is available in the current JSON outputs.

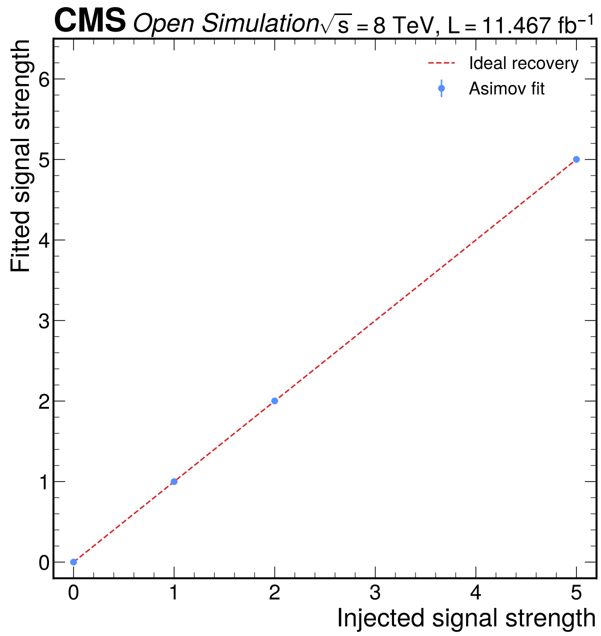

Figure 15: Signal-injection recovery. This validation figure documents the available injected-signal recovery study. The final baseline result remains low sensitivity, but the injection artifact shows that the statistical machinery can recover injected signals within the tested setup.

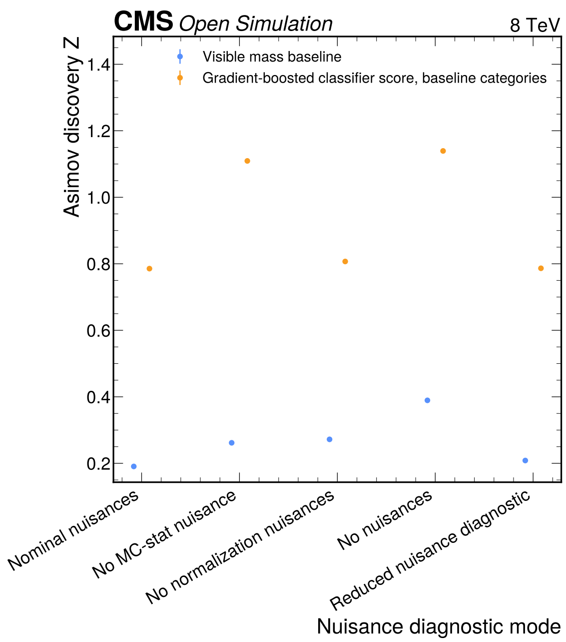

Figure 16: Nuisance audit. This figure summarizes the available nuisance-audit artifact from the inference workflow. It is retained as supporting fit-health context, while the main text reports the final baseline nuisance values from JSON.

10 Final Result

The final result is the visible-mass baseline only. Its profile fit gives

$$\hat{\mu} = 0.4382^{+4.9443}_{-0.4382}, \tag{19}$$

with the lower side bounded at zero, and the 95% CLs upper limit is

$$\mu < 10.7645 \quad (95\%\ \mathrm{CL}_s), \tag{20}$$

compared with a median expected limit of 10.7375. The discovery diagnostic is $Z = 0.0949$ with p-value 0.4622; it is consistent with the low-sensitivity limit interpretation.

Quantity	Final visible-mass result
Best-fit signal strength	0.4382 +4.9443 -0.4382
Observed 95% CLs limit	mu < 10.7645
Expected 95% CLs limit, −2 sigma	mu < 5.4271
Expected 95% CLs limit, −1 sigma	mu < 7.4538
Expected 95% CLs limit, median	mu < 10.7375
Expected 95% CLs limit, +1 sigma	mu < 15.7354
Expected 95% CLs limit, +2 sigma	mu < 22.4849
q0 diagnostic Z	0.0949
Combined data/background	0.9812
Combined chi2/ndf	0.7224

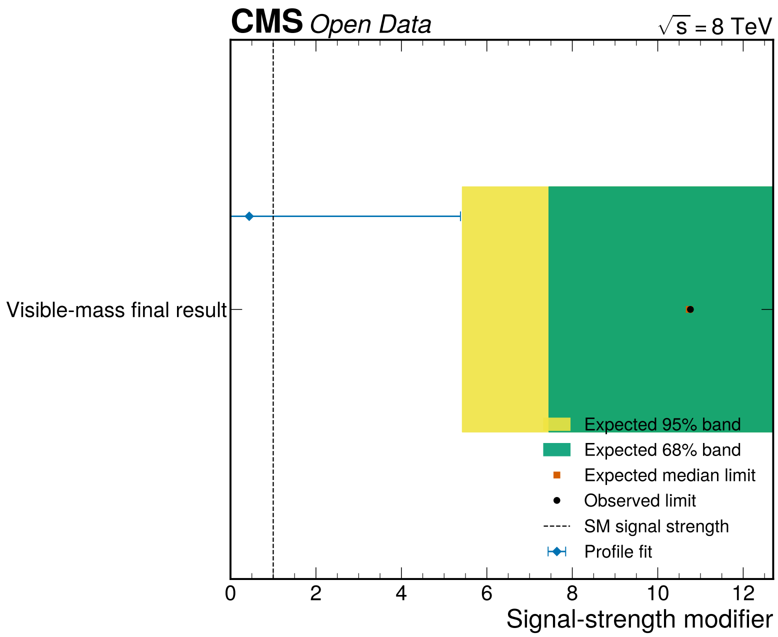

Figure 17: Final baseline limit summary. The figure shows only the validated visible-mass final result with the expected CLs band, observed limit, and profile best-fit interval. No optimized-score value appears in the headline final-result plot.

11 Rejected Classifier Diagnostic

The optimized calibrated-score model is not a final result. It is retained here because it explains why the analysis does not use the best expected-only sensitivity model. The model has median expected limit 1.9735, but on full observed data it gives `mu = 38.3802 +7.0346 -6.1157`, observed limit `mu < 50.0000`, and q0 diagnostic `Z = 12.4698`.

The category fits also indicate model breakdown rather than category-level consistency. The zero-jet category reaches the scan boundary, and VBF/boosted fitted values are several times the Standard Model strength. These values are not used in the abstract, final-result table, conclusion, or published-context comparison.

Diagnostic	Value	Interpretation
Expected median 95% CLs limit	mu < 1.9735	Would be sensitive if observed modelling passed
Observed 95% CLs limit	mu < 50.0000	Hits scan boundary; diagnostic failure
Best-fit signal strength	38.3802 +7.0346 -6.1157	Far outside reduced-analysis plausibility
q0 diagnostic	Z = 12.4698	Rejected, not a discovery claim
Combined data/background	0.6631	Gross normalization mismatch
Combined chi2/ndf	2.4808	Shape/normalization validation diagnostic
Validation gate	fail	The optimized classifier is rejected if it has a boundary-like limit, a large q0 diagnostic, or a gross combined normalization mismatch.

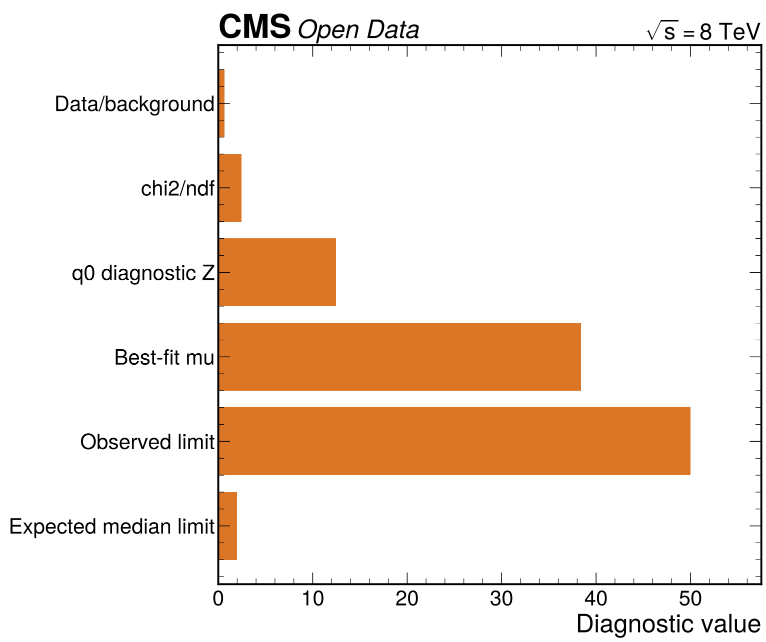

Figure 18: Rejected classifier diagnostic summary. The figure collects the failed optimized-score validation metrics in one diagnostic plot. It is intentionally separated from the final-result section so the failed model is not mistaken for this analysis's physics result.

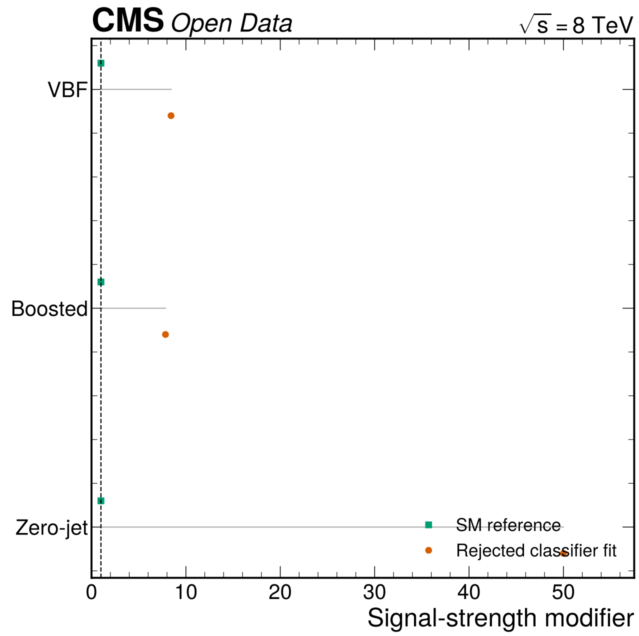

Figure 19: Rejected classifier category fits. The figure shows the single-category signal-strength diagnostics for the optimized classifier model. The large fitted values, including boundary behavior, are evidence for rejection of the classifier result rather than evidence for Higgs production.

12 Comparison With Published Context

Published CMS and ATLAS+CMS H to tau tau analyses use more channels, fuller calibrations, embedded backgrounds, and substantially richer systematic programs than this reduced Open Data study (CMS Collaboration 2014, 2018; ATLAS and CMS Collaborations 2016). The comparison is therefore interpretive, not a validation target. The final result is compatible with published signal strengths only in the weak sense that its uncertainty and upper limit are too large to distinguish $\mu = 1$ from background.

The pull of the final best fit relative to the CMS Run 1 $\mu = 0.78 \pm 0.27$ value is not meaningful as a precision comparison because this analysis has an asymmetric interval with a lower physical boundary. Using the upward uncertainty as a conservative scale gives $(0.438 - 0.78)/4.944 = -0.07$ standard deviations. Relative to the ATLAS+CMS global $\mu = 1.09 \pm 0.11$, the same conservative comparison gives $(0.438 - 1.09)/4.944 = -0.13$ standard deviations. These small pulls reflect low resolving power, not precision agreement.

Result	Scope	Signal-strength or limit	Interpretation
This analysis	8 TeV reduced mu tau_h visible-mass fit	mu = 0.4382 +4.9443 -0.4382; 95% CLs mu < 10.7645; median expected 95% CLs mu < 10.7375	Low-sensitivity Open Data upper limit
CMS Run 1	7+8 TeV multi-channel	mu = 0.78 ± 0.27; observed 3.2 sigma	Evidence-level publication analysis
CMS 2018	13 TeV multi-channel	mu about 1.09; observed 4.9 sigma	Observation in 2016 data
CMS combined in 2018 paper	CMS 7+8+13 TeV	observed 5.9 sigma	CMS observation context
ATLAS+CMS Run 1	7+8 TeV global Higgs combination	global mu = 1.09 ± 0.11	Precision Higgs-rate context
PDG/HXSWG SM	Standard Model normalization	BR(H to tau tau) about 6.3%	Defines the `mu = 1` convention

The validated result cannot exclude or discover a Standard Model-strength Higgs signal in this reduced setup. Its expected limit near 10.74 says that even a perfect background-like observation would only constrain signals roughly an order of magnitude above the SM rate.

13 Limitations

The most important limitation is missing reduced-sample content. No embedded $Z \to \tau\tau$, diboson, single-top, QCD multijet MC, W4/inclusive W+jets, associated-Higgs, or H to WW reduced files are available in the localized mirror used here. These absences limit both sensitivity and systematic realism.

The second limitation is the lack of source-by-source baseline impact refits in the current machine-readable outputs. The final note reports implemented nuisance values, fitted nuisance pulls, and validation metrics, but it does not invent per-source impacts. A future Phase 4 extension should produce an impact JSON by fixing or shifting each nuisance in the final baseline workspace and recording the change in `mu` and the CLs limit.

The third limitation is validation coverage. The final baseline passes the available combined and category checks, but no full observed-data GoF toy ensemble is stored. The VBF category has a residual local tension and the zero-jet chi2/ndf is low. Both effects should be revisited with fuller control-region coverage before making stronger physics claims.

14 Reproduction Contract

The final documents are generated from the current JSON artifacts and pyhf workspaces. A reader with the repository can rebuild the package through pixi:

Command	Purpose
`pixi run phase2-explore`	Reproduce sample inventory and exploration artifacts
`pixi run phase3-all`	Rebuild selected events, selection figures, and sensitivity artifacts
`pixi run phase4a-all`	Rebuild expected-only workspaces and validation
`pixi run phase4b-all`	Rebuild 10% validation artifacts
`pixi run phase4c-all`	Rebuild full-data observed artifacts
`pixi run phase5-docs`	Regenerate this note, PRL draft, references, and Phase 5 figures
`pixi run lint-plots`	Run deterministic plot-standard linting
`pixi run build-pdf`	Rebuild the final analysis note PDF from markdown

The machine-readable numerical sources for the final result are:

Quantity	Artifact
Final fit and validation	baseline result JSON
Final workspace	`pyhf_workspace_baseline_visible.json`
Visible-mass yields	`baseline_visible_yields.json`
Final role metadata	`observed_results.json`
W high-mT scale	`wjets_highmt_scale_full.json`
VBF background scale	`vbf_background_scale.json`
QCD/fake estimates	`qcd_sideband_estimates.json`
Category diagnostics	`category_mu_comparison.json`

15 Appendix A: Bin-Level Baseline Yields

The tables below expose the bin-level values used in the regenerated visible-mass validation plots. They are copied from `baseline_visible_yields.json` and allow the reader to reproduce the ratios in Figure 9, Figure 10, and Figure 11.

15.1 VBF Bin Yields

Visible mass bin [GeV]	Data	Background	QCD/fake	Signal	Data/background
0-60	32	39.59	5.828	0.420	0.808
60-80	23	19.42	5.032	0.808	1.184
80-100	4	14.04	3.154	0.297	0.285
100-120	5	5.84	0.000	0.063	0.856
120-160	6	8.24	0.000	0.002	0.728
160-250	8	9.14	3.733	0.000	0.875

15.2 Boosted Bin Yields

Visible mass bin [GeV]	Data	Background	QCD/fake	Signal	Data/background
0-60	971	974.62	135.863	1.818	0.996
60-80	561	687.29	67.623	3.898	0.816
80-100	243	256.01	35.400	2.596	0.949
100-120	155	146.36	28.825	0.785	1.059
120-160	149	172.98	29.470	0.039	0.861
160-250	104	127.14	28.396	0.035	0.818

15.3 Zero-Jet Bin Yields

Visible mass bin [GeV]	Data	Background	QCD/fake	Signal	Data/background
0-60	3559	3549.55	731.797	2.560	1.003
60-80	3475	3527.74	408.490	6.047	0.985
80-100	772	729.65	179.425	4.651	1.058
100-120	276	279.06	97.238	1.073	0.989
120-160	245	259.12	113.731	0.005	0.945
160-250	124	111.69	38.964	0.005	1.110

16 Appendix B: Validation Bin Ratios and Pulls

Category	Bin ratios	Pulls
VBF	0.808, 1.184, 0.285, 0.856, 0.728, 0.875	-0.78, 0.56, -2.66, -0.31, -0.72, -0.29
Boosted	0.996, 0.816, 0.949, 1.059, 0.861, 0.818	-0.03, -1.25, -0.32, 0.34, -0.84, -1.07
Zero-jet	1.003, 0.985, 1.058, 0.989, 0.945, 1.110	0.02, -0.11, 0.44, -0.07, -0.38, 0.57

17 Appendix C: Numeric Consistency Statement

The final AN and PRL quote baseline numbers from JSON fields only. The best-fit value and profile interval come from `observed_fit.profile_mu_interval` in the baseline result JSON; the CLs limits come from `observed_fit.observed_upper_limit`; validation metrics come from `validation_summary`; and visible-mass bin yields come from `baseline_visible_yields.json`. The optimized classifier values appear only in Section 11 and the corresponding diagnostic JSON/figure.

18 Appendix D: Supporting Validation Figures

This appendix collects additional artifact-backed figures that support the final model choice and reproduce the analysis chain from input objects to the retained visible-mass fit. These figures are not used to quote additional physics results; they document why the final result is intentionally limited to a weak visible-mass CLs limit and why more aggressive observables are not promoted to the headline result.

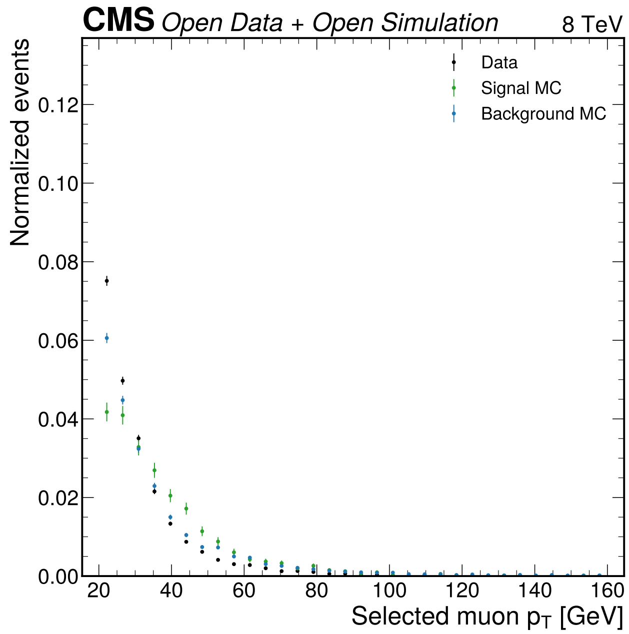

Figure 20: Muon transverse-momentum preselection comparison. This figure checks the reconstructed muon leg in the selected reduced samples. The agreement is not used as a standalone calibration, but it supports the use of the common muon selection in all final categories.

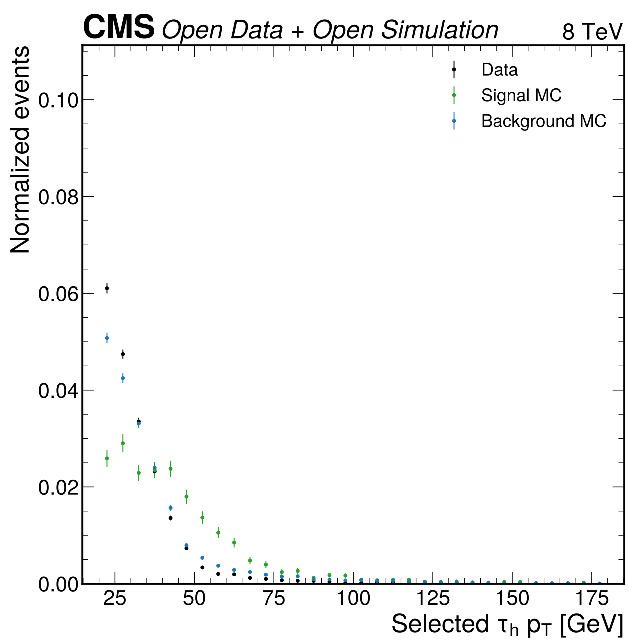

Figure 21: Hadronic-tau transverse-momentum preselection comparison. This figure checks the tau_h leg after the loose public-object acceptance and tight anti-muon requirement. The final analysis assigns a tau/open-data acceptance nuisance because the public reduced sample does not include the full trigger and tau scale-factor program.

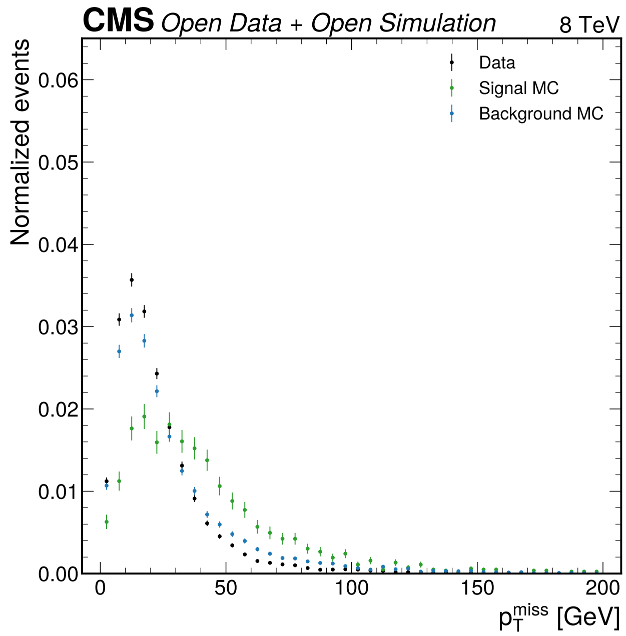

Figure 22: Missing transverse momentum comparison. This figure checks the missing-momentum observable before it enters the transverse-mass and add-MET cross-check definitions. It motivates treating add-MET based observables as diagnostics rather than replacing the validated visible-mass result.

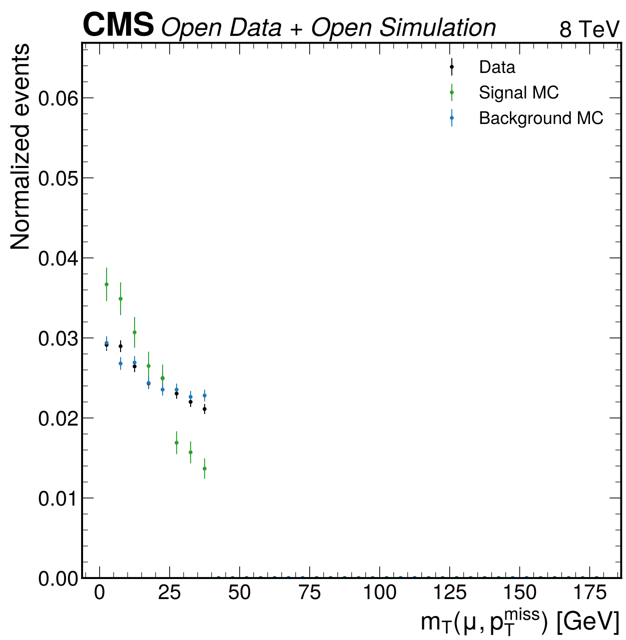

Figure 23: Muon-MET transverse-mass comparison. This distribution defines the signal and W-control regions used by the final workflow. The separation between the low-mT signal region and high-mT control region underlies the W+jets normalization factor in Equation 5.

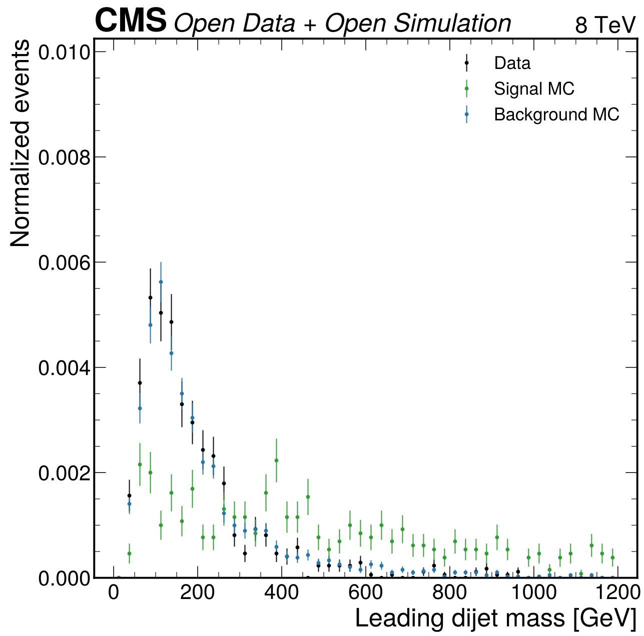

Figure 24: VBF dijet-mass comparison. This figure checks the dijet mass observable that enters the VBF category definition. It supports the category split while also showing why the VBF region remains statistically limited in the reduced sample set.

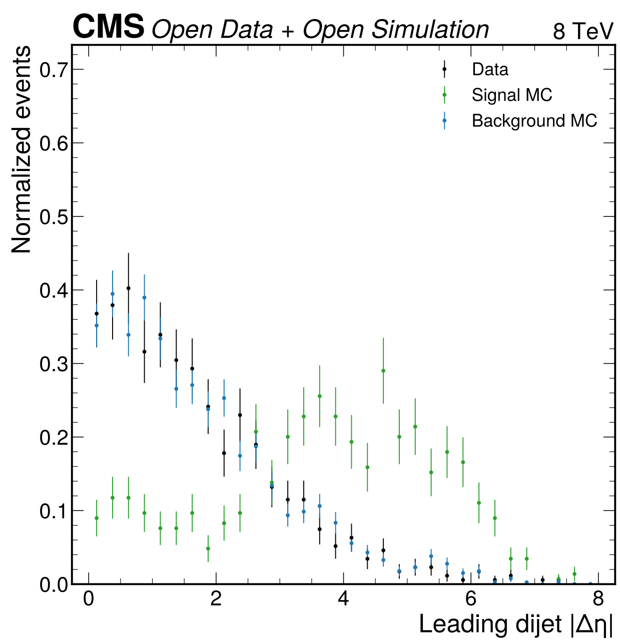

Figure 25: VBF dijet-rapidity-separation comparison. This figure checks the second VBF topology variable used in category assignment. Together with Figure 24, it documents the topology basis for separating VBF from boosted and zero-jet events.

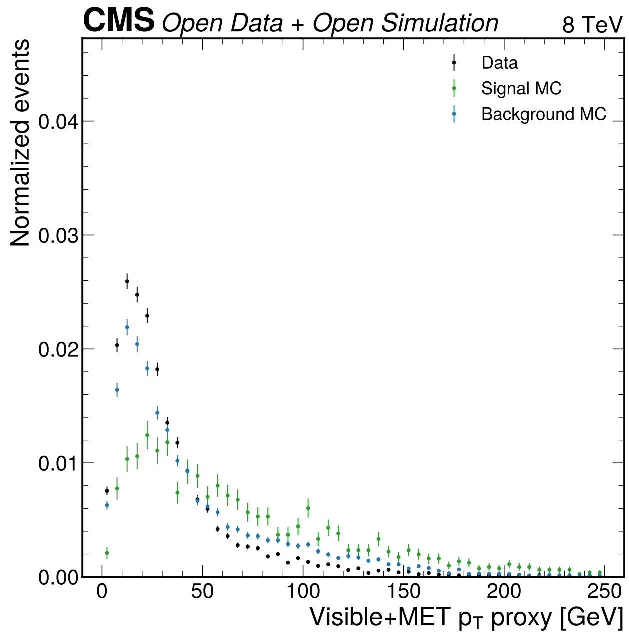

Figure 26: Visible transverse-momentum proxy comparison. This figure summarizes the reconstructed di-tau transverse-momentum proxy available in the reduced inputs. It is retained as a modelling cross-check, not as a final-fit observable.

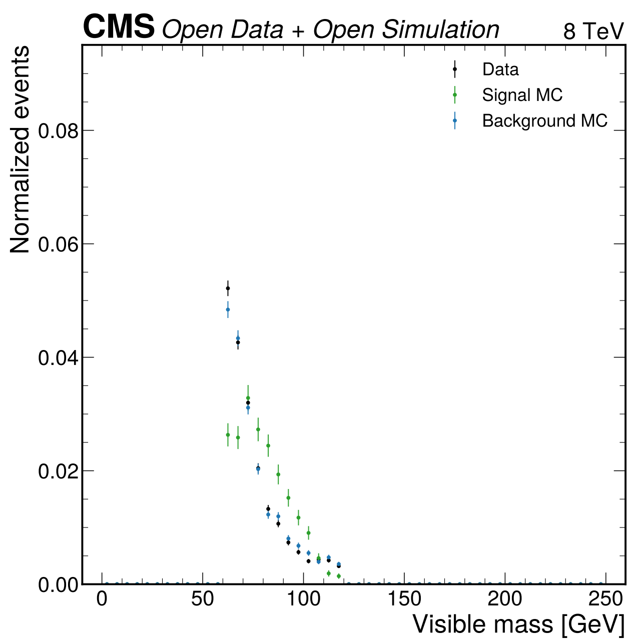

Figure 27: Z-rich visible-mass validation. This control-style figure checks the visible-mass shape in a Z-enriched region. The final note uses it as supporting evidence for the visible-mass baseline while keeping the DY/Z normalization uncertainty explicit.

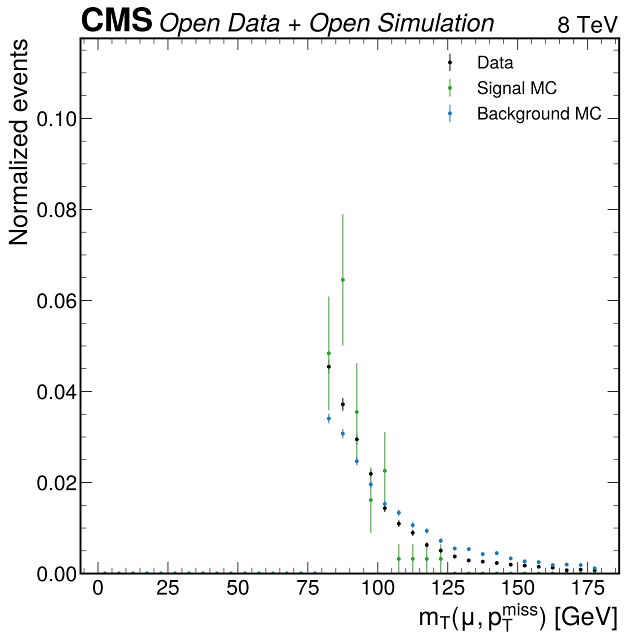

Figure 28: W high-mT control comparison. This figure shows the control-region distribution used for the W+jets normalization correction. It complements the numerical W scale in Equation 5 by showing the region from which the scale is derived.

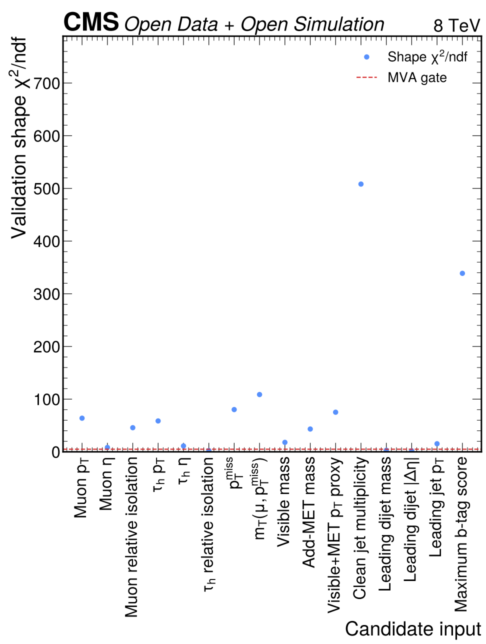

Figure 29: MVA input modelling summary. This figure records the input-modelling chi2 diagnostics that prevented an unqualified promotion of a classifier observable. It is included to make the rejected-classifier decision reproducible without reading the Phase 3 code.

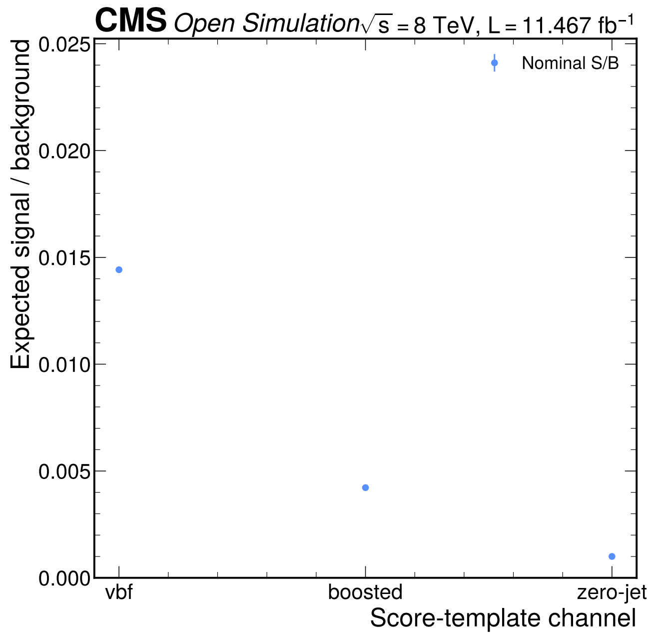

Figure 30: Expected signal-over-background summary. This expected-stage diagnostic shows why alternative observables were worth studying: the classifier and category choices can improve expected separation. The observed-data validation failure, not the expected-only sensitivity, determines the final model role.

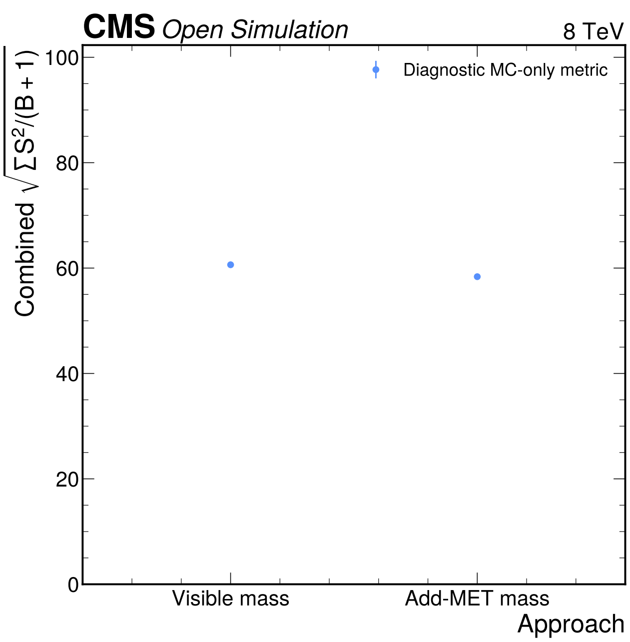

Figure 31: Approach comparison summary. This figure compares the broad observable strategies considered in the analysis chain. It documents that the final visible-mass result is a validation-driven choice rather than the most aggressive expected-sensitivity option.

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